

Technical training.
Product information.

F39 Complete Vehicle



BMW Service

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Technical Training

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General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:



Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left-hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as a result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

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The information contained in this document forms an integral part of the BMW Group Technical Qualification and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

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F39 Complete Vehicle

1. Introduction

The F39 will be the first in the new BMW X2 series. It unifies the sporty form of a BMW Coupé with the robust character of a BMW X. The model stands for sportiness and activity. The F39 will be launched in the 1st quarter of 2018 and will be available with all-wheel drive.

Many systems and components of the F39 are identical to the F48 X1. This Technical Reference Manual focuses on changes and new features compared to F48 X1 in addition to the technical data.



BMW X2 – F39

1.1. Models

One model will be available at the start of production:

- BMW X2 xDrive28i

Model	Engine	Power output [kW (HP)]	Torque [Nm (lb-ft)]	Displacement [cm]
BMW X2 xDrive28i	B46A20O0	170 (228)	350 (258)	1998

A second model will follow later in 2018:

- BMW X2 sDrive28i

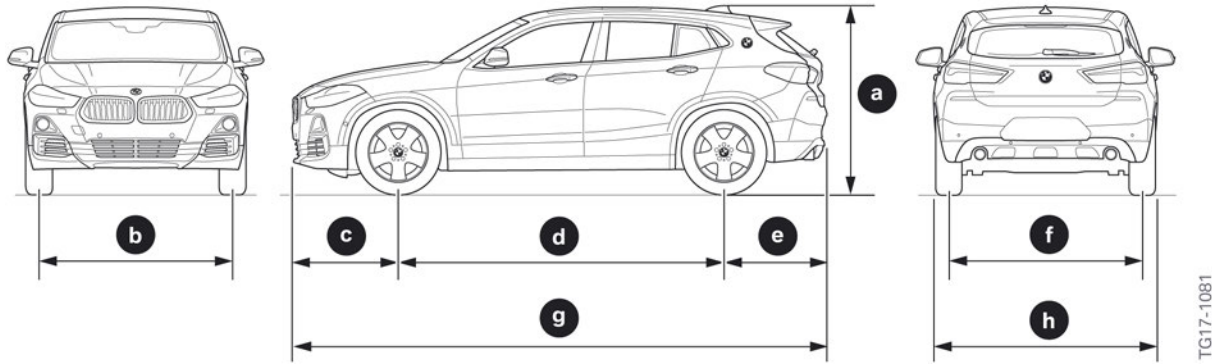
Model	Engine	Power output [kW (HP)]	Torque [Nm (lb-ft)]	Displacement [cm]
BMW X2 sDrive28i	B46A20O0	170 (228)	350 (258)	1998

F39 Complete Vehicle

1. Introduction

1.2. Dimensions

The following section shows the outer dimensions of the F39. The following table compares the dimensions of the F39 with those of an F48:



F39 outer dimensions

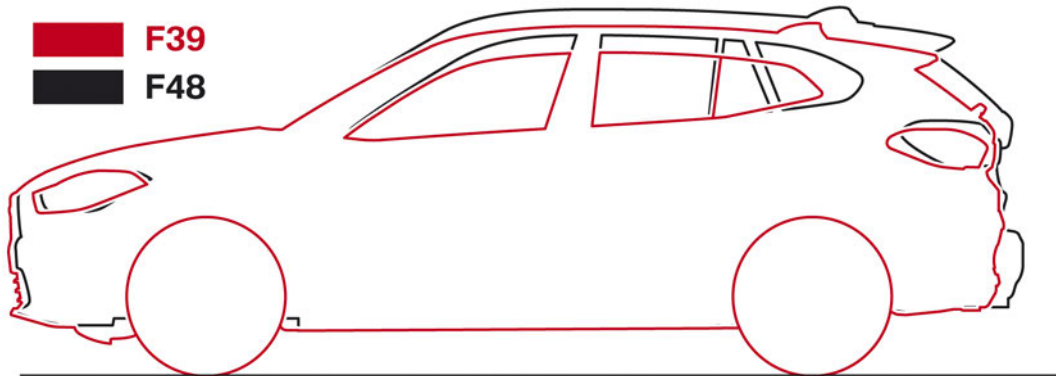
Index	Explanation		F39	F48	Difference
a	Vehicle height with production add-on parts	[mm] (inches)]	1543	1612	-69 (-2.7)
b	Front track width, basic wheels	[mm] (inches)	1561	1561	±0
c	Front overhang	[mm] (inches)	854	842	+12 (+0.5)
d	Wheelbase	[mm] (inches)	2670	2670	±0
e	Rear overhang	[mm] (inches)	836	927	-91 (-3.5)
f	Rear track width, basic wheels	[mm] (inches)	1562	1562	±0
g	Vehicle length	[mm] (inches)	4360	4439	-79 (-3.1)
h	Width excluding exterior mirrors	[mm] (inches)	1824	1821	+3 (+0.1)

F39 Complete Vehicle

1. Introduction

1.3. Silhouette comparison

A silhouette comparison between the F48 and the F39 shows the differences in the dimensions of the two vehicles.



TG17-1080

Comparison of an F39 silhouette with an F48 silhouette

1.4. Weights and load capacities

The following table compares the weights and payloads of the F39 and the F48, using the BMW "X2 xDrive28i" and the "X1 xDrive28i" with an automatic transmission:

		F39 X2 BMW xDrive28i Automatic transmission	F48 X1 BMW xDrive28i Automatic transmission
Vehicle curb weight	[lbs]	3662	3659
Gross weight	[lbs]	4729	4720
Payload	[lbs]	900	900

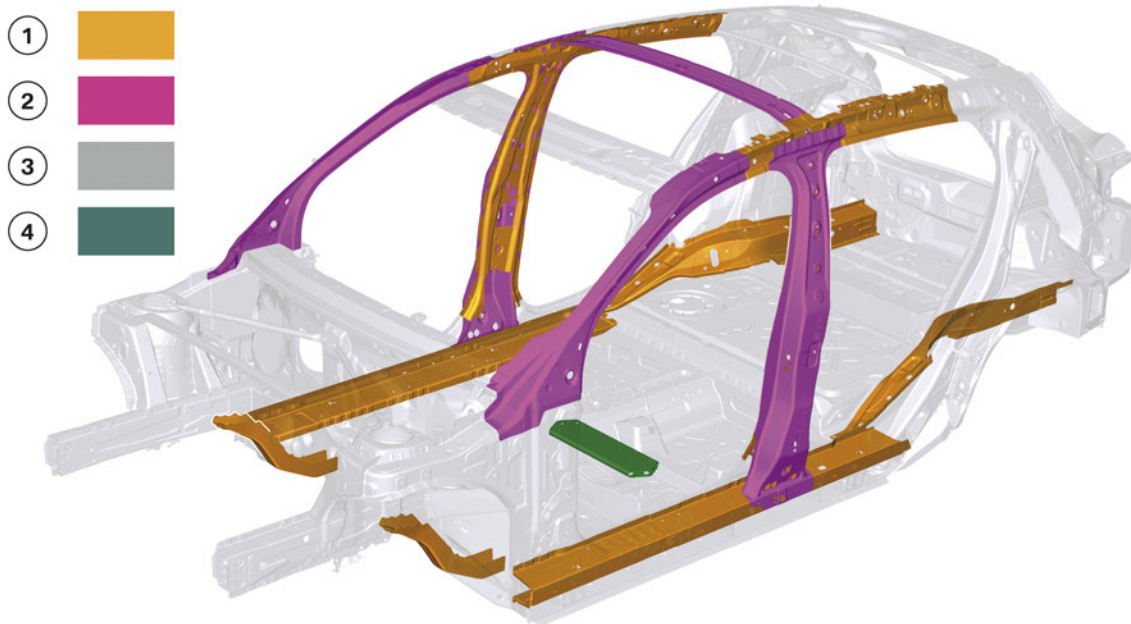
F39 Complete Vehicle

2. Body

2.1. Material

2.1.1. Body structure

The multi-phase steels and hot-formed steels ensure maximum strength of the passenger cell with low weight, thus making a huge contribution to passive safety.



TK17-1093

F39 body structure materials

Index	Explanation
1	Multi-phase steels (> 300 N/mm), 10% of the body weight
2	Heat-shaped steels (> 900 N/mm), 6% of the body weight
3	Other steels, 84% of the body weight
4	Aluminium, <1% of the body weight

Hot-formed steel is used in the reinforcements of the A- and B-pillars, for example. It is only used for safety-relevant components in the body structure.

Due to their high carbon equivalent, hot-formed steels have limited weldability. This means that they can only be repaired by one or more special joining methods, such as by welding, bonding and riveting.

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2. Body

2.2. Luggage compartment

2.2.1. Volume

The luggage compartment of the F39 has a capacity of (21.6 ft³). By folding down the rear seat backrests, the luggage compartment capacity can be expanded to (50.1 ft³) and converted to a flat cargo area. The cargo space is 15% smaller than the cargo space in the F48. The rear seat backs can be folded with the 40:20:40 split.



F39 Luggage compartment capacity

Index	Explanation
A	Luggage compartment with the rear seat backrests not folded down
B	Luggage compartment with the rear seat backrests folded down

F39 Complete Vehicle

2. Body

2.2.2. Luggage compartment removable panel

The F39 features a storage compartment below the luggage compartment removable floor panel. In addition to the existing 4 lashing eyes, another 2 lashing eyes and a load area net are installed in the luggage compartment with the “Storage package” as standard.



F39 Luggage compartment removable panel

A storage compartment with luggage compartment net and a 12 V connection in the right side trim panel are standard equipment. The removable panel can be folded and removed. Under the removable panel, the vehicle can optionally be equipped with an Space-saver spare tire and jack (OE 300).

2.3. C-pillar BMW Roundel

A BMW Roundel is located on both C-pillars of the F39. The Roundel conjures up BMW Coupés of days gone by and underscores the sporting character of the BMW X2. The Roundel is bonded to the vehicle body.

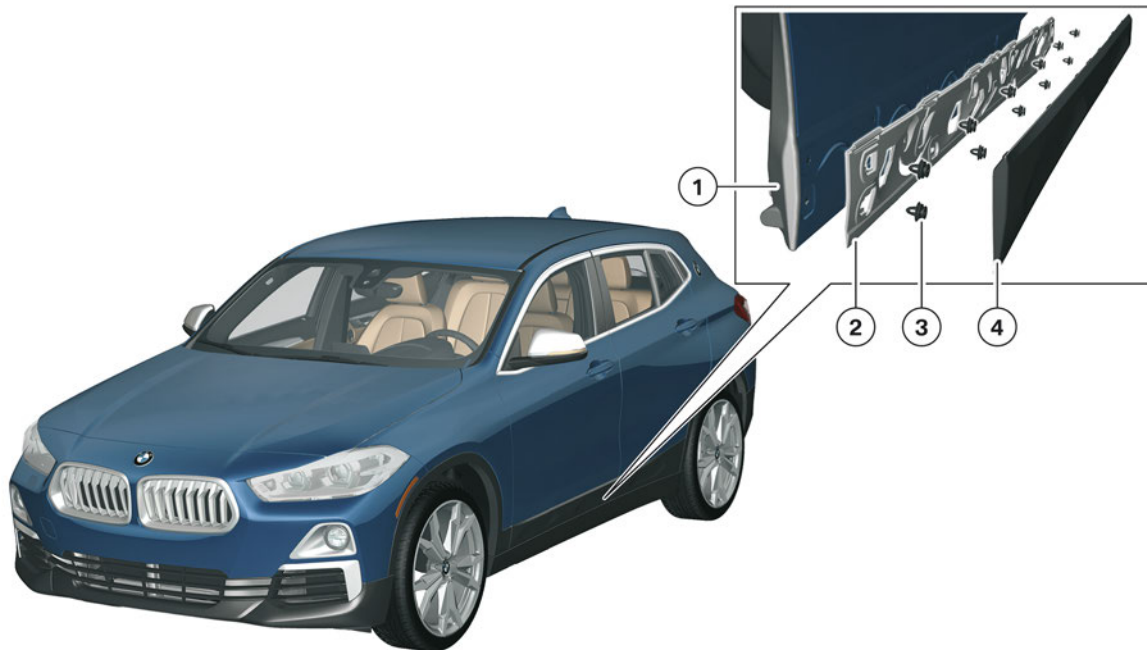


F39 Complete Vehicle

2. Body

2.4. Cladding

The F39 features a colored band on all sides, the so-called cladding. It has a different color depending on the line.



TK17-1658

F39 Cladding – line basis

Index	Explanation
1	Front left door
2	Support
3	Clip
4	Top shell

The top shell is connected to the vehicle body by means of a support. The support is attached to the vehicle body with clips.

F39 Complete Vehicle

2. Body

2.5. Tailgate

The rear of the F39 can be opened using the BMW Roundel on the tailgate. An automatic operation of the tailgate is standard equipment for the F39.



F39 Tailgate

In conjunction with the optional Comfort Access Keyless Entry, the tailgate can also be opened and closed contactlessly by means of targeted foot movement. Two sensors detect the movement contactlessly via a capacitive measurement.



F39 Smart opener

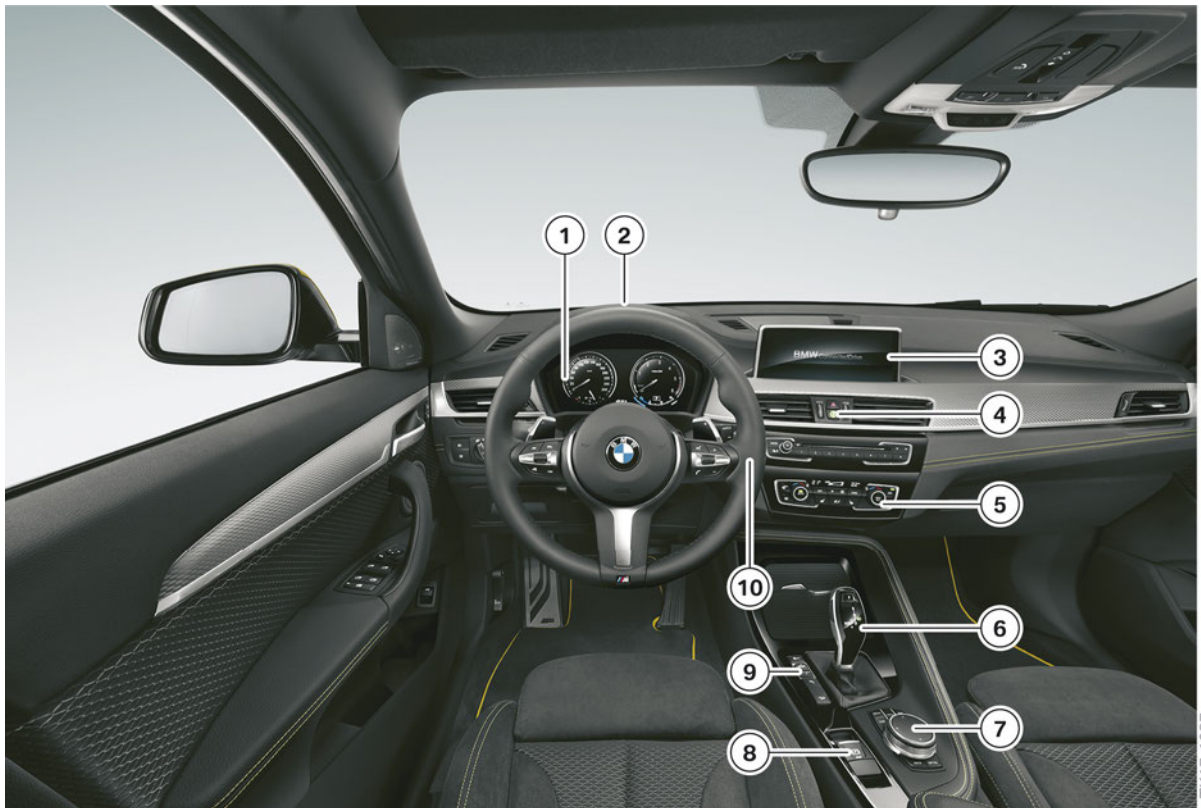
F39 Complete Vehicle

2. Body

2.6. Interior equipment

2.6.1. Overview

The following graphic provides an overview of the display and operating elements of the F39:



F39 Cockpit M SportX package

Index	Explanation
1	Instrument cluster (KOMBI)
2	Head-Up Display (HUD)
3	Central Information Display (CID)
4	Intelligent Safety button
5	Radio operating unit/heating and air conditioning operating unit
6	Gear selector switch
7	Controller (CON)
8	Parking brake button
9	Driving experience switch
10	Multifunction steering wheel

F39 Complete Vehicle

2. Body

2.6.2. Feature packages

In addition to the comprehensive offering of optional equipment, the F39 can also be individualized with the following equipment packages: The equipment packages contain both general optional equipment and equipment-specific features. Sport front seats are standard on both equipment packages.

Two equipment packages are available with the new BMW X2 :

- Basic Package
- M SportX Package



F39 Exterior equipment packages

Index	Explanation
A	Basic Package
B	M SportX Package

F39 Complete Vehicle

2. Body



F39 Interior equipment packages

Index	Explanation
A	Basic Package
B	M SportX Package

F39 Complete Vehicle

2. Body

2.6.3. Seat bench

A continuous rear seat bench without forward/back adjustment is always installed in the F39. The seats can be folded down individually and have two latch positions for enlarging the luggage compartment space.



F39 Seat bench

Index	Explanation
A	Rear seat bench with the backrests not folded down
B	Rear seat bench with the middle backrest folded down
C	Rear seat bench with all backrests folded down

F39 Complete Vehicle

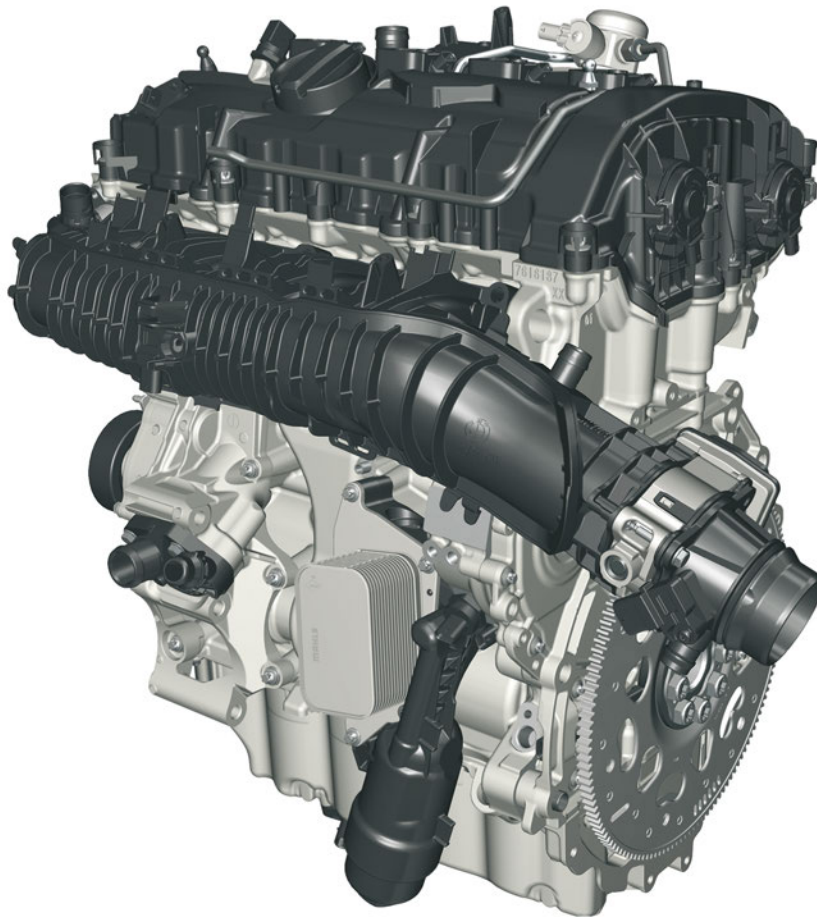
3. Engine

3.1. Gasoline engine

Model	Engine	Power output [kW (HP)]	Torque [Nm (lb-ft)]
BMW X2 xDrive28i	B46A2000	170 (228)	350 (258)

3.1.1. B46 engine

Technical data



TA17-1071

Engine B46A2000

Parameters	Unit	B46A2000
Displacement	cm	1998
Number of cylinders	—	4

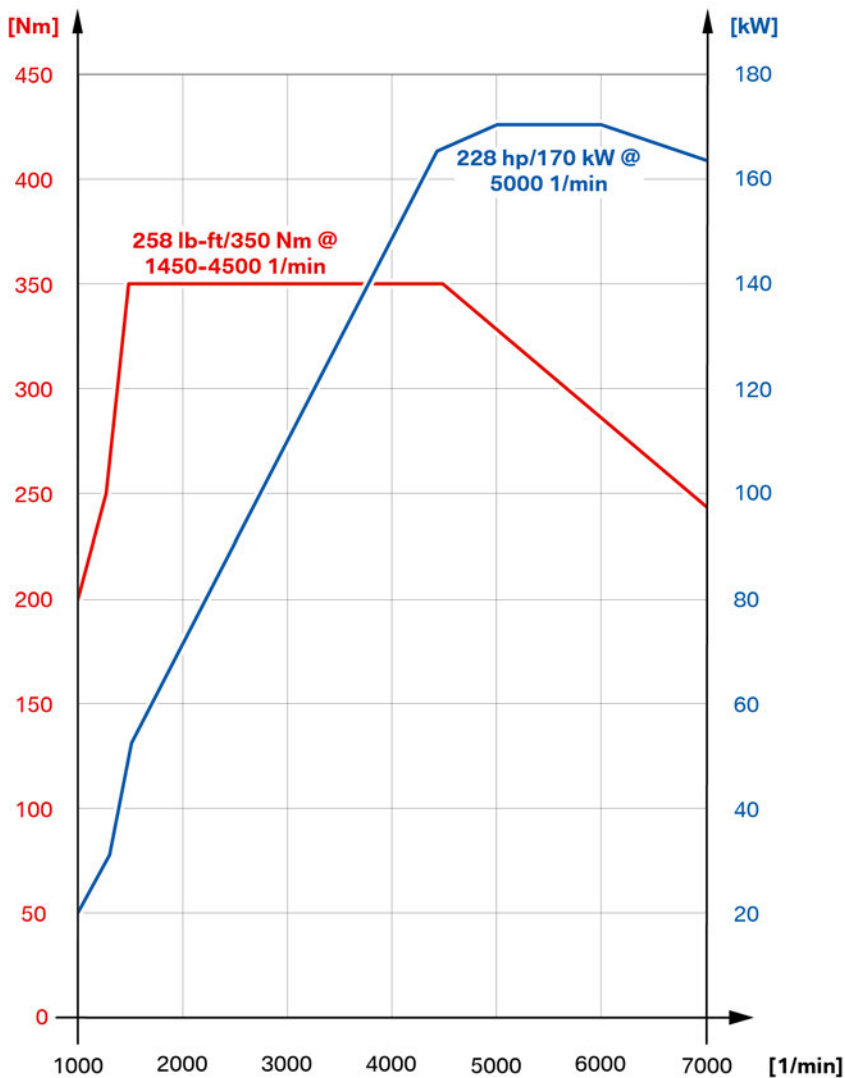
F39 Complete Vehicle

3. Engine

Highlights

- SULEV injectors
- Catalytic converter close to engine
- DME MEVD 8.4.

Full-load diagram for B46A2000



TA17-1070_2

Full-load diagram for B46A2000

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4. Transmission

The F39 is equipped with an AISIN GA8F22AW 8 speed transmission. The gear selector with a cable is used like in the F48 X1.



F39 Gear selector X2 F39

4.1. Overview

Model	Engine	Automatic transmission
BMW X2 xDrive28i	B46A20O0	AISIN

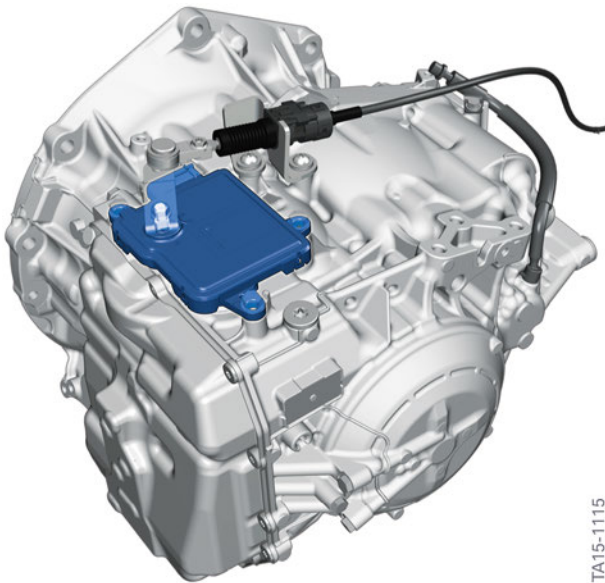
F39 Complete Vehicle

4. Transmission

4.2. Automatic transmission

4.2.1. Overview

In the F39, the AISIN automatic transmission (GA8F22AW) is used.



F39 Automatic transmission

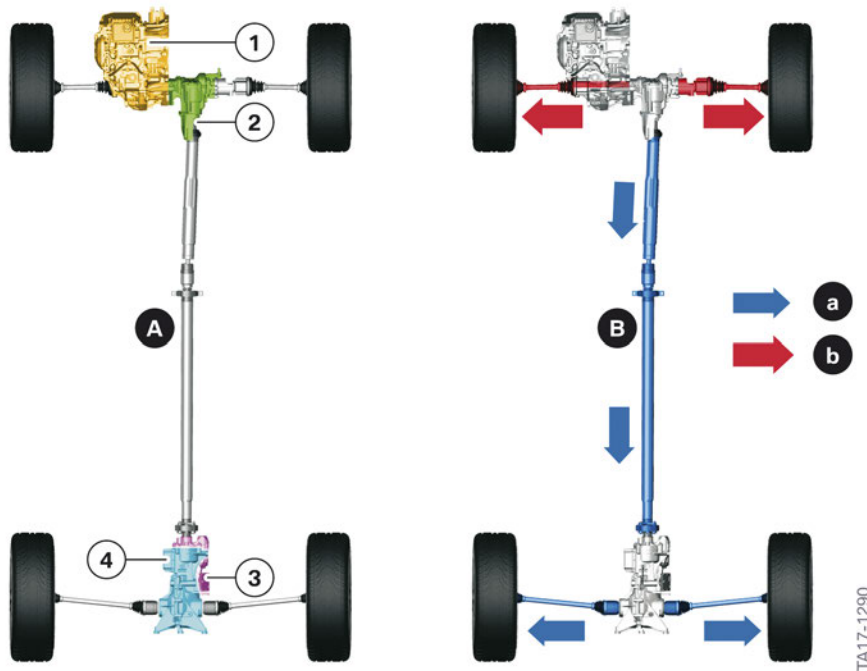
Further information on the AISIN automatic transmission (GA8F22AW) can be found in the ST1508 "F48 Powertrain" Technical Reference Manual.

F39 Complete Vehicle

4. Transmission

4.3. Four-wheel drive

The all-wheel drive system in F39 is based on a front-wheel drive design and is already in use in series F48 BMW X1.



F39 4-wheel drive

Index	Explanation
A	Component overview
B	Power flow
1	Automatic transmission
2	Bevel gears
3	Longitudinal torque distribution with electro-hydraulic clutch (integrated in rear axle)
4	Rear axle final drive
a	Driving power to the rear wheels
b	Driving power to the front wheels

Further information is available in the ST1508 "F48 Complete Vehicle" Technical Reference Manual.

F39 Complete Vehicle

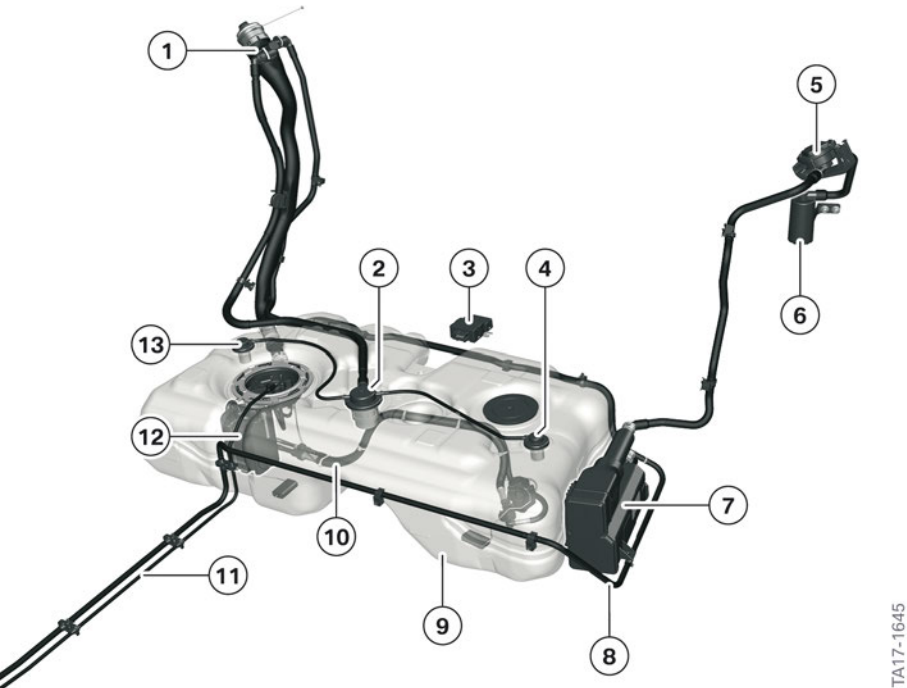
5. Fuel Supply

5.1. Fuel tank

The F39 features a fuel tank with a capacity 16.1 gallons.



F39 Fuel tank



F39 Fuel tank 16.1 gallon filling capacity

Index	Explanation
1	Fuel filler neck
2	Filling limiter 1
3	Fuel pump control electronics
4	Filling limiter 2
5	Natural Vacuum Leak Detection (NVLD)
6	Dust filter
7	Carbon canister

F39 Complete Vehicle

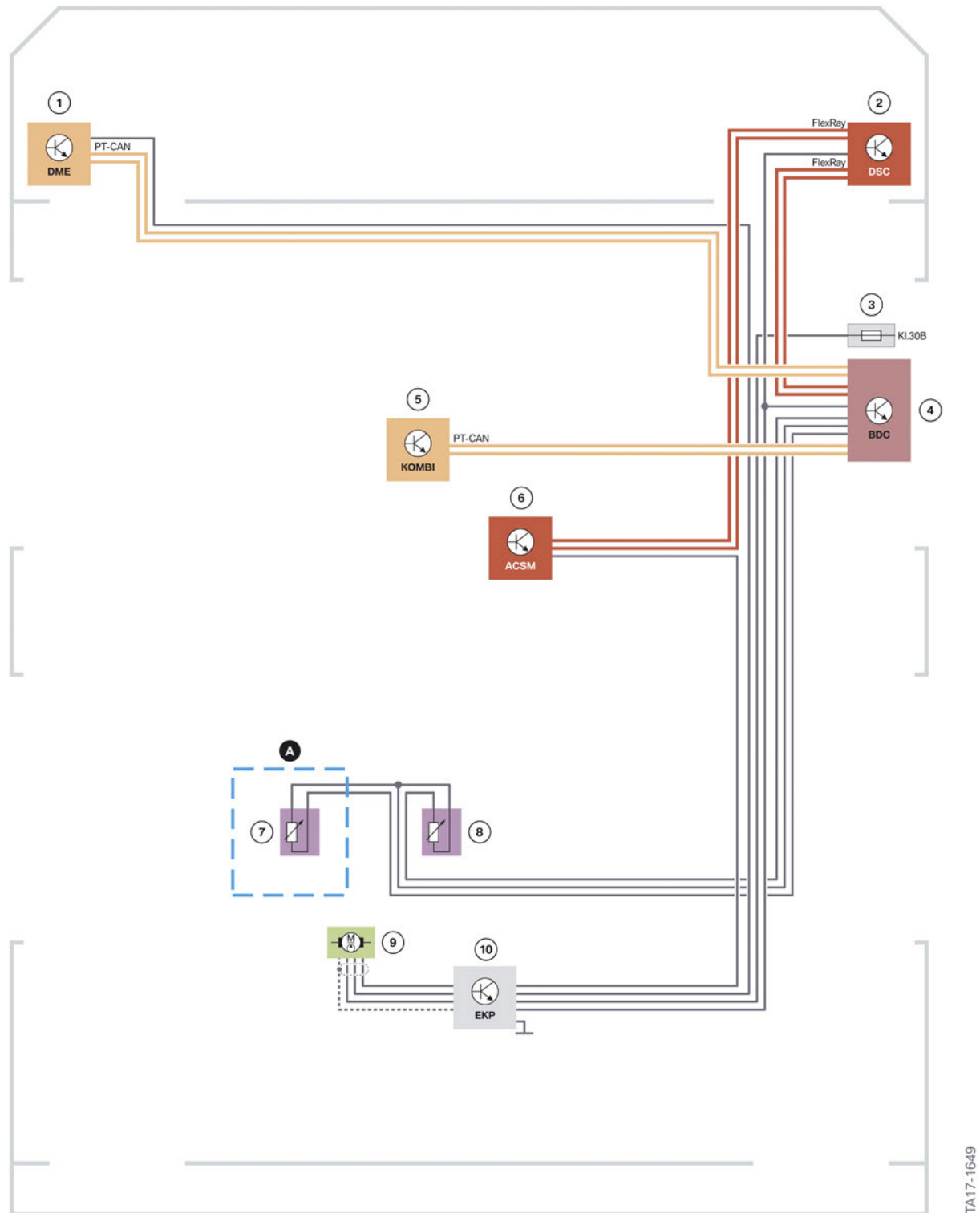
5. Fuel Supply

Index	Explanation
8	Purge air line
9	16.1 gallon fuel tank
10	Fuel pump line to side of fuel tank
11	Feed line
12	Electric fuel pump
13	Filling limiter 3

F39 Complete Vehicle

5. Fuel Supply

5.2. System wiring diagram



F39 System wiring diagram for the fuel supply

F39 Complete Vehicle

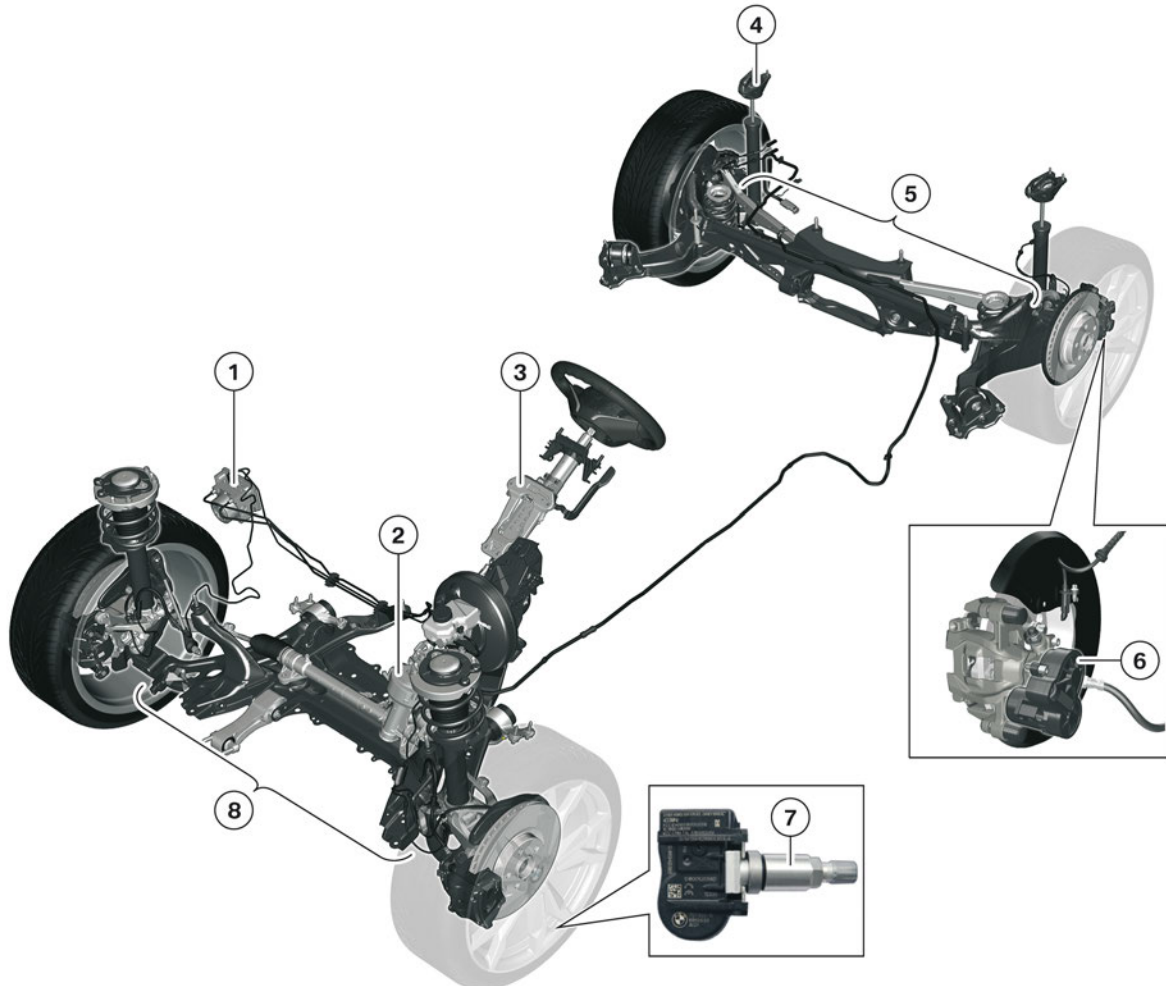
5. Fuel Supply

Index	Explanation
A	Fuel tank with filling capacity of 16.1 gallons
1	Engine control unit (DME)
2	Dynamic Stability Control (DSC)
3	Front passenger power distribution box
4	Body Domain Controller (BDC)
5	Instrument cluster
6	Advanced Crash Safety Module (ACSM)
7	Left filling level sensor
8	Right filling level sensor
9	Electric fuel pump
10	Control unit, electric fuel pump

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6. Chassis and Suspension

6.1. Overview



TG17-1079

F39 Overview of chassis and suspension

Index	Explanation
1	Dynamic Stability Control (DSC)
2	Electronic Power Steering (EPS)
3	Mechanically adjustable steering column
4	Shock absorber, rear
5	Multi-link rear axle
6	Rear floating caliper brake with electric parking brake actuator
7	Wheel electronics for tire pressure control
8	Single-joint spring-strut-type front axle

F39 Complete Vehicle

6. Chassis and Suspension

The principle of the single-joint spring-strut-type front axle was adopted from F48. The F39 has specific axle kinematics for minimizing drive influences on the steering and increased rigidity for agile drivability.

The use of aluminium an (swivel bearing) and high-strength steels used for the front axle support and wishbone, reduce the unsprung masses. The friction-optimized wheel bearings help reduce carbon dioxide emissions.

The modular multi-link rear axle was adopted from the F48.

In addition, Dynamic Damper Control is included in the Dynamic Handling Package (OE ZDH) that also includes M Sports Steering that can be ordered as optional equipment .

The suspension on the F39 was modified in the following areas, unique suspension bushings, modified sway bars and specific alignment settings. This is different than the F48 suspension to give the F39 a slightly different ride and sportier handling.

6.2. Tire Pressure Monitor System

The F39 is fitted with tire pressure control as part of the standard equipment. The function is integrated in the Dynamic Stability Control (DSC).

6.2.1. Notes for Service

The service life of the batteries of the wheel electronics is roughly 10 years or 300,000 km (186,000 miles). The current value of the service life counter, stated in years and months, can be read out with the help of the diagnosis system.



When removing/installing the wheel electronics, the following points must be observed:

- When fitting tires, make sure the sensor is in the correct position as specified in the repair instructions.
 - The tire pressure monitor system will malfunction if the incorrect version of the wheel electronics is installed.
 - Do not use high pressure cleaners to clean the wheel rim with integrated wheel electronics when the tire has been removed.
 - The wheel electronics need to be renewed if tire sealant was used.
 - The valve and valve seat must be thoroughly cleaned before the wheel electronics are installed.
 - The wheel electronics must not be treated with solvents or cleaning agents.
 - The wheel electronics must not be blown out with compressed air.
 - The wheel electronics may only be cleaned by wiping them down with a clean cloth.
-

F39 Complete Vehicle

6. Chassis and Suspension

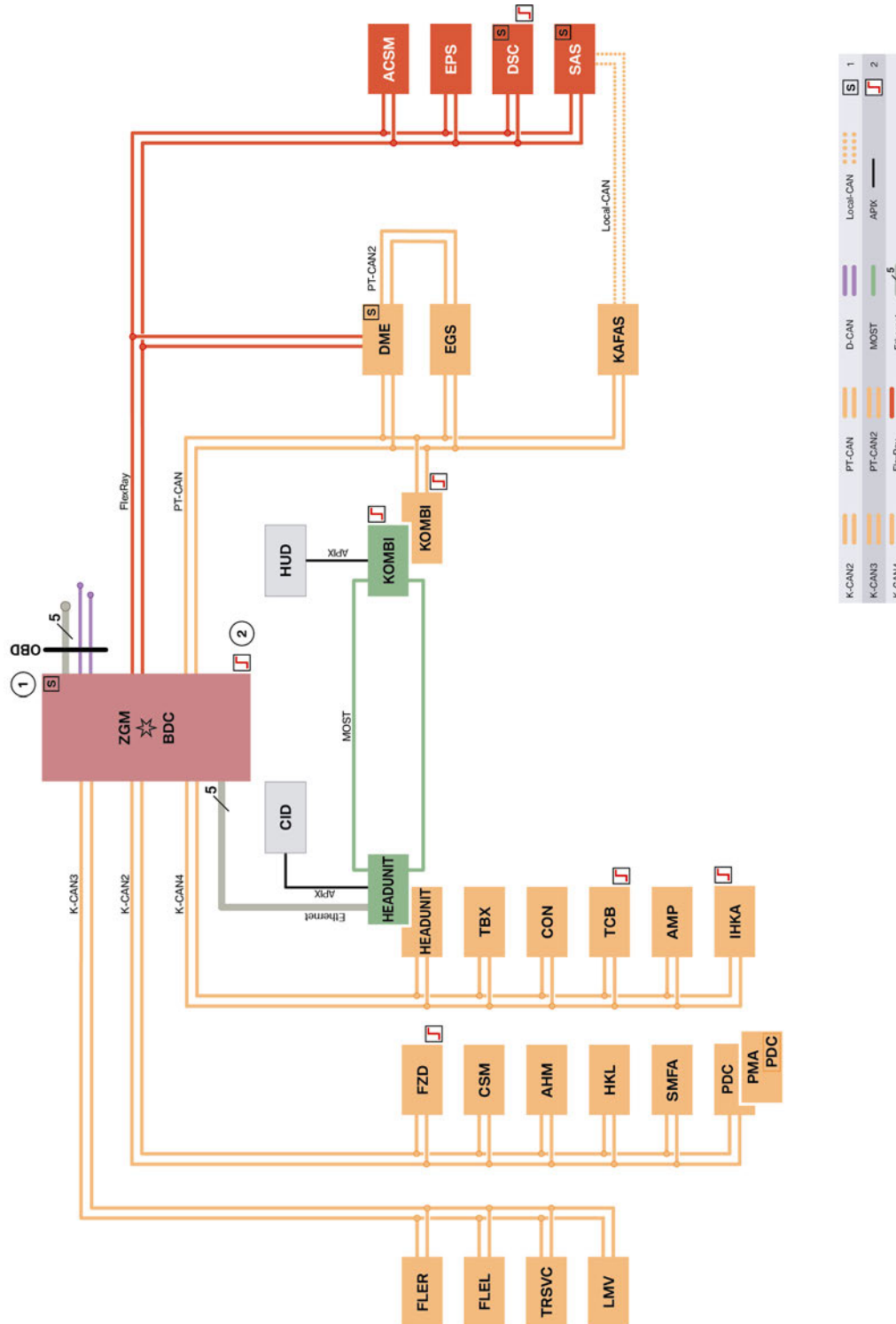
6.3. Further information

System	Further information
Electronic Power Steering (Electromechanical Power Steering) (EPS)	ST1508 "F48 Chassis and Suspension" Technical Reference Manual
Electromechanical parking brake	ST1508 "F48 Chassis and Suspension" Technical Reference Manual
Tire Pressure Monitor	ST1508 "F48 Chassis and Suspension" Technical Reference Manual
Dynamic Damper Control	ST1508 "F48 Chassis and Suspension" Technical Reference Manual
Dynamic Stability Control (DSC)	ST1508 "F48 Chassis and Suspension" Technical Reference Manual

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7. General Vehicle Electronics

7.1. Bus overview



F39 bus overview

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7. General Vehicle Electronics

Index	Explanation
1	Start-up node control units for starting and synchronizing the FlexRay bus system
2	Control units with wake-up authorisation
ACSM	Advanced Crash Safety Module
AHM	Trailer module
AMP	Audio amplifier
BDC	Body Domain Controller
CID	Central Information Display
CON	Controller
CSM	Car Sharing Module
DME	Digital Motor Electronics
DSC	Dynamic Stability Control
EGS	Electronic transmission control
EPS	Electronic Power Steering
FLEL	Frontal Light Electronics Left
FLER	Frontal Light Electronics Right
FZD	Roof function center
HEAD UNIT	Head unit
HKL	Automatic operation of tailgate
HUD	Head-Up Display
IHKA	Integrated automatic heating / air conditioning
KAFAS	Camera-based driver support systems
KOMBI	Instrument panel
LMV	Longitudinal torque distribution
PDC	Park Distance Control
PMA	Parking Manoeuvring Assistant
SAS	Optional equipment system
SMFA	Seat module, driver
TBX	Touchbox
TCB	Telematic Communication Box
TR SVC	Control unit for rear view camera and Side View
ZGM	Central Gateway Module

F39 Complete Vehicle

7. General Vehicle Electronics

7.2. Exterior lights

7.2.1. Headlights

In the F39, LED headlights are standard.

LED headlights with Cornering Lights	
Equipment	Standard
Design	
Bulbs	
LED	• Side lights • Daytime lights • Low-beam headlight • High-beam headlight • Turn indicator
Additional functions	• Cornering lights

7.2.2. Fog light

The F39 is supplied with LED fog lights.



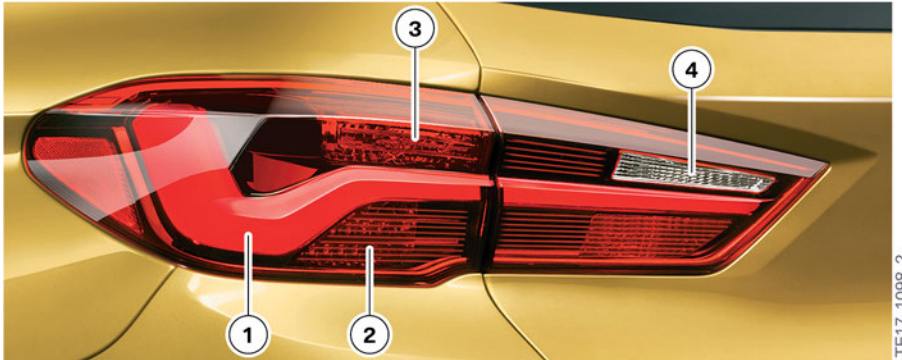
F39 Fog light

F39 Complete Vehicle

7. General Vehicle Electronics

7.2.3. Rear lights

The brake and tail lights of the rear lights of the F39 are equipped with LEDs. All other functions are implemented with bulbs.



F39 Rear light

Index	Explanation
1	Rear light (LED)
2	Brake light (LED)
3	Turn indicator
4	Reversing light

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8. Assistance Systems

8.1. Overview

The assistance systems of the F39 are identical with those of a F48. This Technical Reference Manual shows only an overview of the available assistance systems and packages.

The following assistance systems are available for the F39:

- Driving Assistant Package (OE 5AS)
- ACC Stop & Go + Active Driving Assistant (OE 5AT)
- Park Distance Control front and rear (OE 508)
- Parking Manoeuvring Assistant (OE 5DP)
- Dynamic Cruise Control with braking function standard equipment
- Rear view camera standard equipment

In the F39, a lot of the optional equipment known from other BMW models is still only available in conjunction with optional equipment packages Driving Assistant Package (OE 5AS) or ACC Stop & Go + Active Driving Assistant (OE 5AT). The following table should show which system is included in the different optional equipment packages:

System	Driving Assistant Package (ZDA) (OE 5AS)	Active Driving Assistant (OE 5AT)
Automatic high beams	X	
Lane departure warning	X	
Speed limit info	X	
City collision mitigation	X	
Forward collision warning	X	
ACC with Stop & Go function		X

Further information on the assistance systems can be found in the ST1508 "F48 Complete Vehicle" Technical Reference Manual.

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9. Displays and Controls

9.1. Head-Up Display

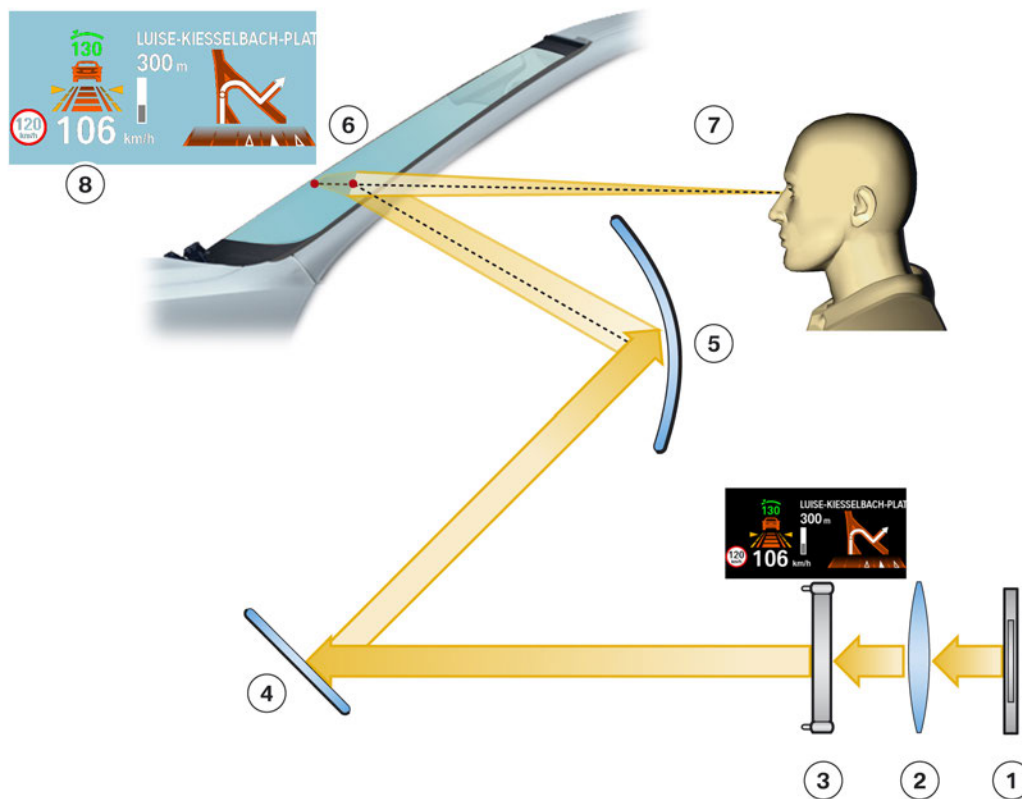
The Head-Up Display (HUD) projects a virtual, transparent image onto the windshield in the driver's field of view.

The BMW Head-Up Display (OE 610) offers the driver the option of viewing all information relevant for driving directly in the field of view, freely floating over the roadway.

The Head-Up Display used in the F39 can project the colors red, green and blue and by mixing these colors enables the projection of image content in all colors of the red/green/blue spectrum. The HUD receives the control signals and the image content from the instrument cluster via an APIX interface. The power supply of the HUD is also effected in the F39 by the instrument cluster.

9.1.1. Functional principle

The Head-Up Display is similar to a projector. A light source is required to project the information. Using 16 LEDs, the information is shown on a color, transparent red/green/blue 1.8" display. The resolution is 480x240 pixels. The image obtained is projected into the area of vision of the driver using several deflection mirrors and the windshield.



TE10-1565

HUD operating principle

F39 Complete Vehicle

9. Displays and Controls

Index	Explanation
1	Light source
2	Honeycomb reflector with diffuser
3	TFT projection display
4	Plane mirror
5	Curved mirror
6	Windshield
7	Observer's point of vision
8	Projected image

The screen appears freely floating over the roadway.

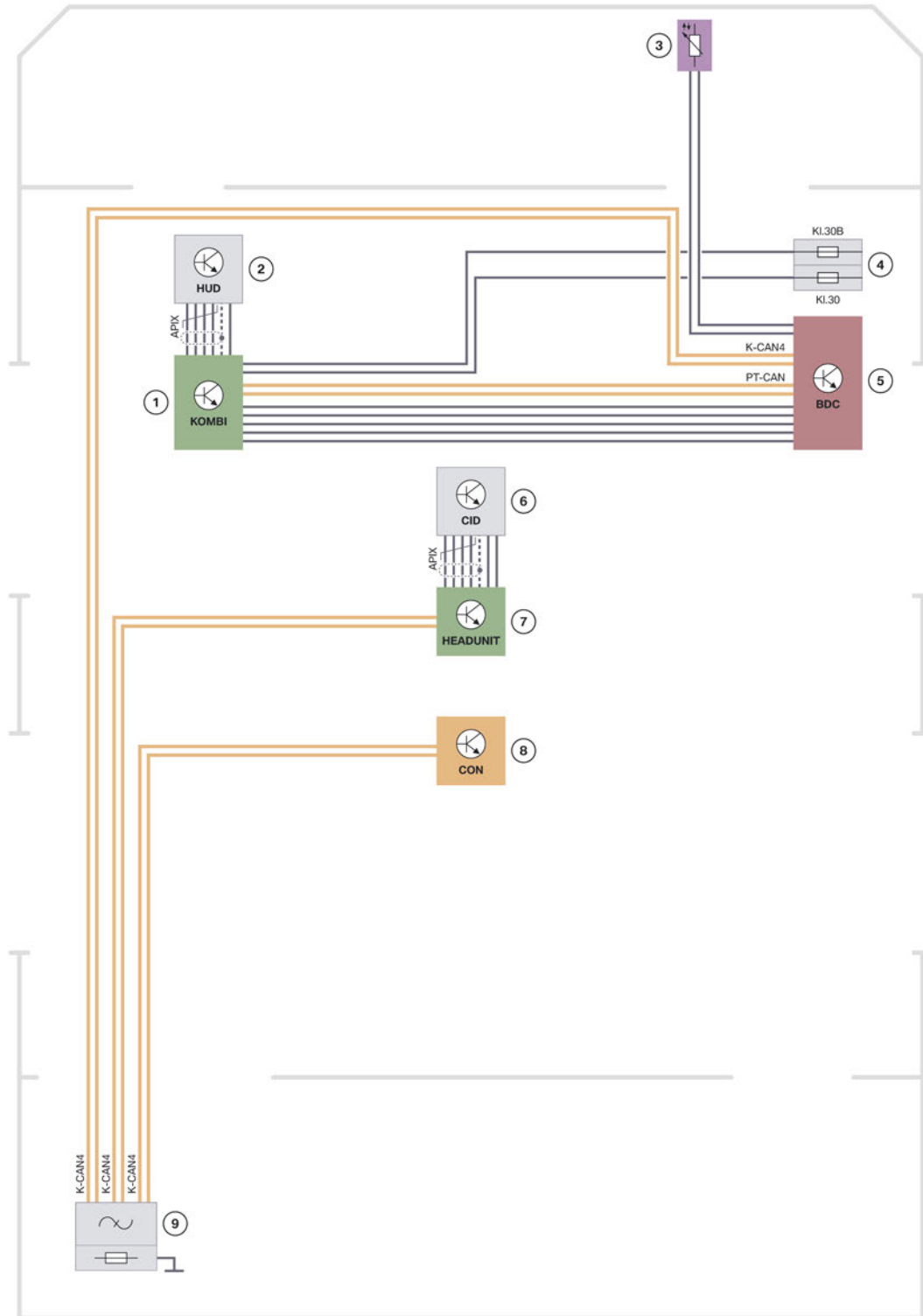
A special windshield is required for the HUD. A wedge-shaped foil in the windshield prevents double images. On standard windshields there is no wedge-shaped foil, double images would be displayed with this type of windshield.

The head-up display is fitted above the steering column, immediately behind the instrument cluster.

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9. Displays and Controls

9.1.2. System wiring diagram



TE14-1584

F39 System wiring diagram for Head-Up Display

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9. Displays and Controls

Index	Explanation
1	Instrument cluster (KOMBI)
2	Head-Up Display (HUD)
3	Outside temperature sensor
4	Power distribution box, passenger compartment
5	Body Domain Controller (BDC)
6	Central Information Display (CID)
7	Head unit
8	Controller (CON)
9	K-CAN terminator

9.1.3. Settings

The following settings can be changed using the controller:

- Head-Up Display ON/OFF
- Brightness of display
- Rotation of display
- Height of display
- Displayed information

9.1.4. Display

The changeover of day design to night design is the same for the display in the instrument cluster.

The following content can be displayed:

- Vehicle speed
- Driver assist warning messages
- Navigation display
- Check Control messages
- Set speed of ACC/DCC
- Speed limit info
- Entertainment lists

With the exception of the current vehicle speed and the driver assistance warning messages, the information displayed can be deselected by the controller.

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9. Displays and Controls

The information displayed in the Head-Up Display is influenced by:

- Seat position
- Objects on the cover of the Head-Up Display
- Sunglasses with certain polarizing filters
- Wet roadway
- Inadequate lighting conditions

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10. Infotainment

10.1. Instrument cluster

The F39 has two instrument clusters, which are known from other BMW models:

- Basic version of instrument cluster
- High version of instrument cluster

①



②



TE13-2686

F39 versions of instrument clusters

Index	Instrument cluster	Display
1	Basic version	2.7"
2	High version	5.7"

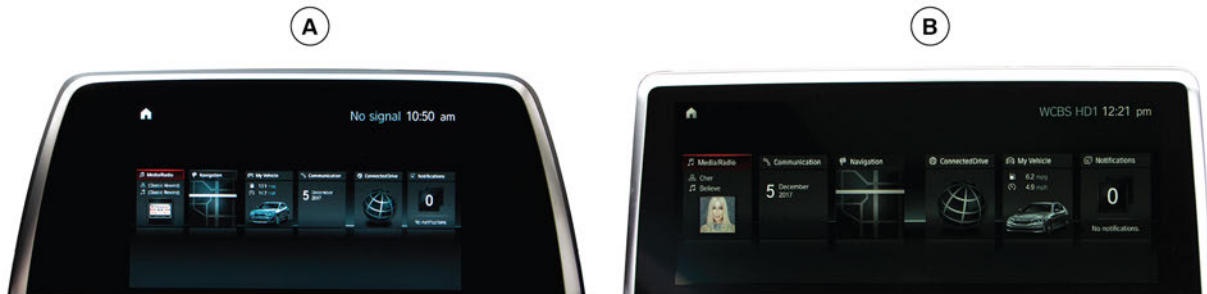
F39 Complete Vehicle

10. Infotainment

10.2. Central information display

10.2.1. Overview

In the F39, two different Central Information Displays (CID) are installed, depending on the equipment. These are a 6.5" display and an 8.8" display.



F39 Central information displays

Index	Explanation
A	CID with 6.5", installed with media system (standard equipment) with optional Navigation Business (OE 6UN) with touch screen. If no navigation installed, CID is not available with touch screen.
B	CID with 8.8", installed with Navigation Plus (OE 6UP) with touch screen

10.2.2. User interface

The F39 features user interface ID6 (BMW iDrive of the 6th generation) already used in the G30 and G12.

With the ID6, up to 6 tiles can be displayed in the main menu simultaneously. The information on the tiles is displayed in real-time and continuously refreshed. Thus, for example, an active navigation can continue to be used. The contents of the following tiles can be customized:

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Menu	Tile setting
Navigation	• Map • Arrow view
On-board computer	• Reduced on-board computer • On-board computer • Driving style analysis
ConnectedDrive	Market specific features. A number of examples are listed: • Weather • News • Local Wiki • Country information

In the F39, two different Central Information Displays (CID) are installed, depending on the equipment. These are a 6.5" display and an 8.8" display.



F39 Central information tile displays

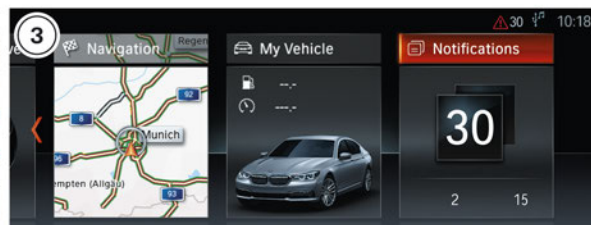
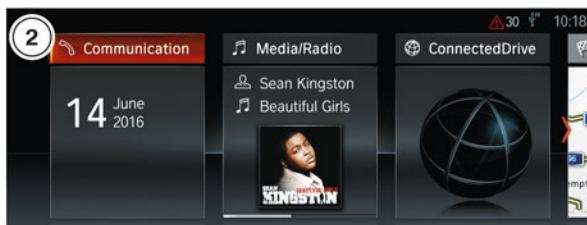
Index	Explanation
A	CID with 6.5", detailed view of the 2 left-hand tiles
B	CID with 8.8", detailed view of the 3 left-hand tiles

The tile setting is made in the "My vehicle" menu or in the main menu by pressing the OPTION button on the controller.

Double-clicking on the menu button on the controller changes the display on the main menu to three larger tiles.

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Detailed view of the tiles

Index	Explanation
1	View of all 6 tiles
2	Detailed view of the 3 left-hand tiles
3	Detailed view of the 3 right-hand tiles

The arrangement of the tiles is not predetermined and can be adjusted individually by touch (OE 6UP or OE 6UN only) or by controller.



Individual arrangement of the tiles

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10.3. Head units

10.3.1. Overview

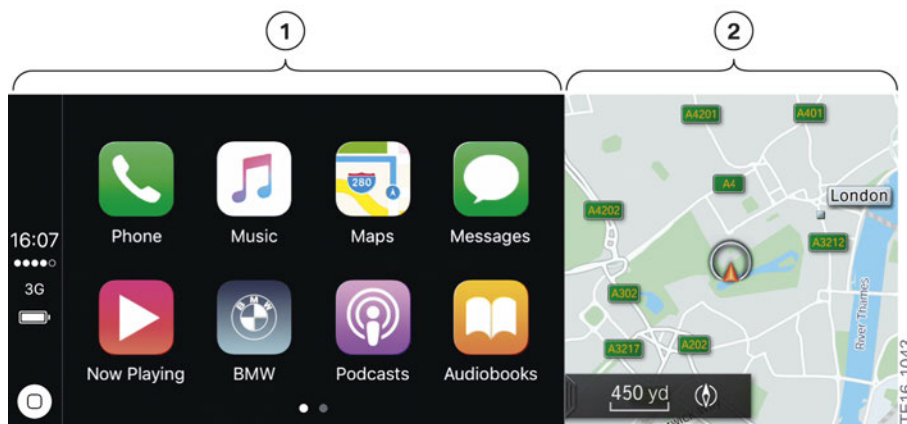
The equipment of the different head units can be found in the following table:

Optional equipment	Head unit	CID	Controllers	Navigation
Media system (basic equipment)	Head Unit Basic	6,5"	5-button	No
Navigation Business (OE 6UN)	Head Unit Basic with navigation	6,5" with touch screen	7-button	Yes
Navigation Plus (OE 6UP)	Head Unit Basic with navigation	8,8" with touch screen	7 buttons with touch controller	Yes

Further information may be found in the ST1508 "F48 Complete Vehicle" Technical Reference Manual.

10.4. Apple® CarPlay

Apple® CarPlay offers the capability of easily and conveniently reproducing selected content from the iPhone® in the vehicle. Displaying the Apple® icons on the Central Information Display (CID) permits intuitive operation and control of the functions.



Apple® CarPlay in the Central Information Display (CID)

Index	Explanation
1	Main screen with Apple® CarPlay
2	Assistance window, here, for example, shown with the BMW navigation map

Apple® CarPlay is displayed only in the main screen. The Apple® CarPlay logo is displayed if there is no function in the assistance window. Apple® CarPlay is activated by pressing and holding the menu button on the controller.

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10.4.1. Prerequisites

The vehicle and the iPhone® must satisfy certain prerequisites before Apple® CarPlay can be used.

The following optional equipment must be installed in the **vehicle** to be able to use Apple® CarPlay:

- OE 6CP (Apple® CarPlay Compatibility)
- OE 6AK (ConnectedDrive Services)
- OE 6UN (Navigation Business) or OE 6UP (Navigation with Touch Controller).

The iPhone® must fulfil the following conditions to be able to use Apple® CarPlay:

- iPhone® 5 and newer
- Operating system iOS 9.3.3 or later.

Selected data attributes are swapped between the smartphone and the vehicle when Apple® CarPlay is used.

10.5. Telephone systems

The F39 can be equipped for hands-free operation and wireless charging:

- Convenient telephone with extended smartphone connection.
- Wireless charging (OE 6NW).

10.5.1. Convenient telephone with extended smartphone connection

The following list provides an overview of the functions offered by convenient telephone with the expanded smartphone connection:

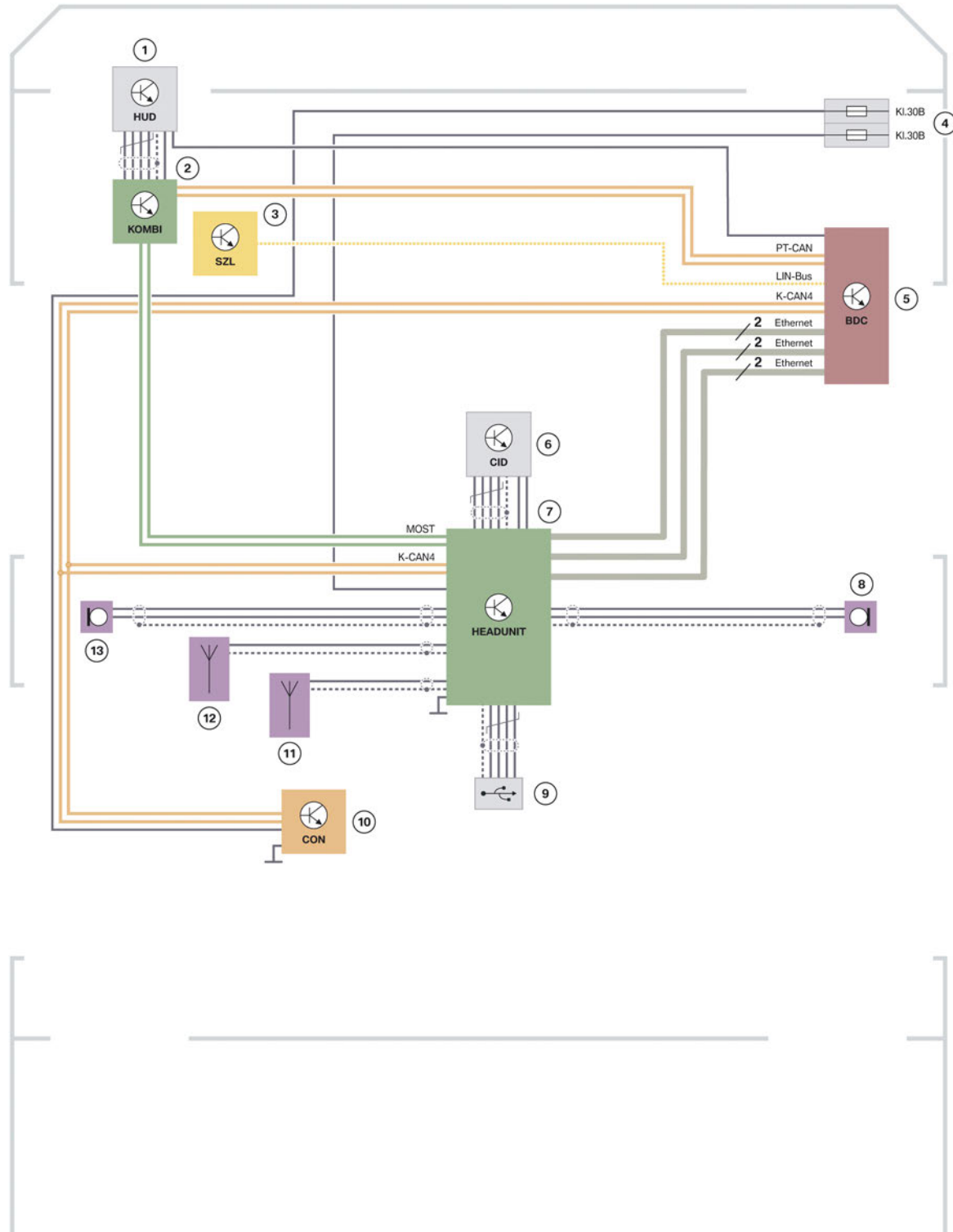
- Two USB interfaces each with 2.1 A charge current
- Simultaneous pairing of 2 mobile phones and an audio device.
- Improved hands-free capability for the front passenger by means of a second microphone
- Hotspot optional equipment (OE 6WD) can be added

No base plate is fitted for the convenient telephone with extended smartphone connection optional equipment. With this equipment it is therefore also not possible to charge a mobile phone through a snap-in adapter. Because the wiring harness connection has been deleted, the system **cannot** be retrofitted for use of a wireless charging station. Connection to the roof-mounted antenna is also no longer possible with this optional equipment.

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System wiring diagram



F39 System wiring diagram for convenient telephone with extended smartphone connection

TE17-1100

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Index	Explanation
1	Head-Up Display (HUD)
2	Instrument cluster (KOMBI)
3	Steering column switch cluster (SZL)
4	Power distribution box, front passenger
5	Body Domain Controller (BDC)
6	Central Information Display (CID)
7	Head unit
8	Microphone 2
9	USB port
10	Controller CON
11	WLAN antenna
12	Bluetooth antenna
13	Microphone 1

10.5.2. Telephone with wireless charging

telephone with wireless charging (OE 6NW) has the same functions as the convenient telephone with extended smartphone connection. Additionally, the smartphone can be charged wirelessly and has a connection to the roof-mounted antenna.

This function is possible only for mobile phones compliant with the Qi[®] standard. The standard is identifiable by the icon. It ensures that the chargers are compatible with the devices.



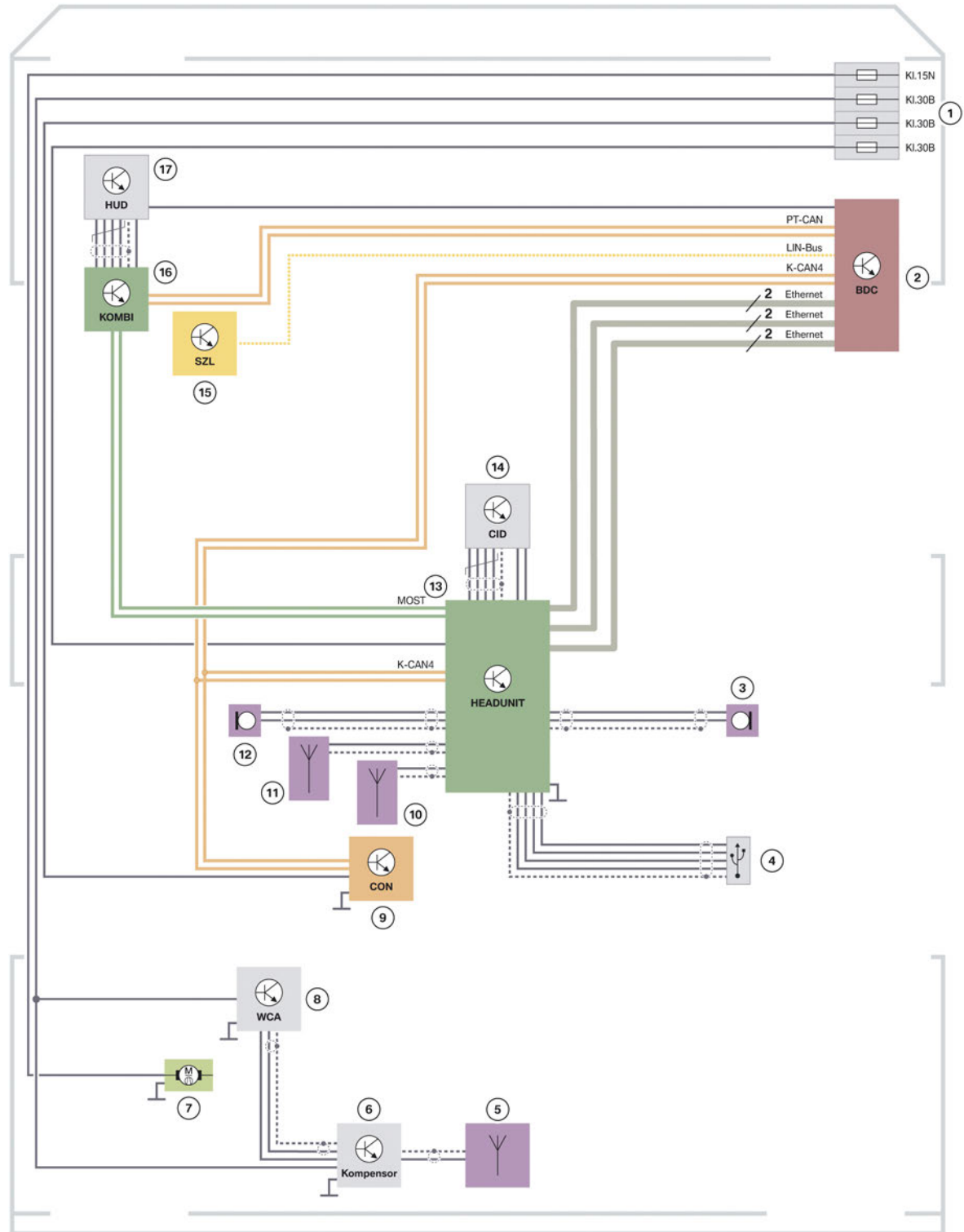
Qi[®] detection

For information on compatibility, go to: www.bmw.com/bluetooth

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System wiring diagram



F39 telephone with wireless charging system wiring diagram

TE17-1102

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10. Infotainment

Index	Explanation
1	Power distribution box, front passenger
2	Body Domain Controller (BDC)
3	Microphone 2
4	USB port
5	Roof antenna
6	Compensator
7	Center armrest storage compartment fan
8	Wireless charging unit
9	Controller CON
10	WLAN antenna
11	Bluetooth antenna
12	Microphone 1
13	Head unit
14	Central Information Display (CID)
15	Steering column switch cluster (SZL)
16	Instrument cluster (KOMBI)
17	Head-Up Display (HUD)

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10.5.3. Wireless charging station



F39 Wireless charging station under the center armrest

Index	Explanation
1	Wireless charging station

In the F39, the wireless charging is implemented using a special wireless charging station. It is located in the center armrest storage compartment.

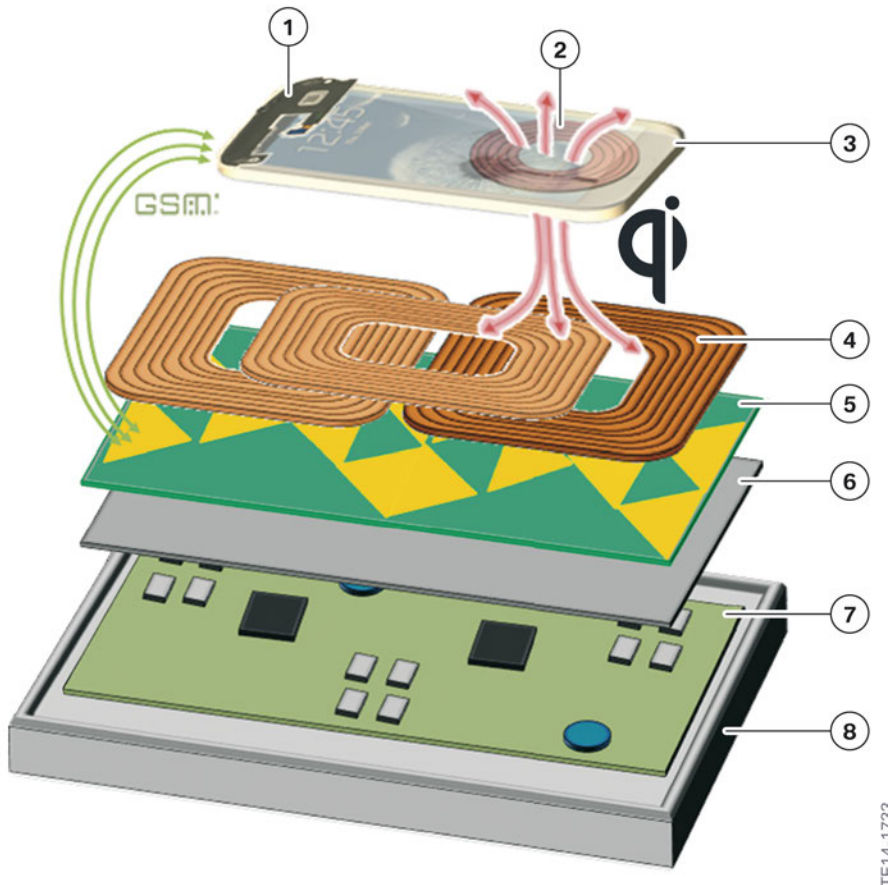


Wireless charging station

F39 Complete Vehicle

10. Infotainment

Function



Basic principle of wireless charging

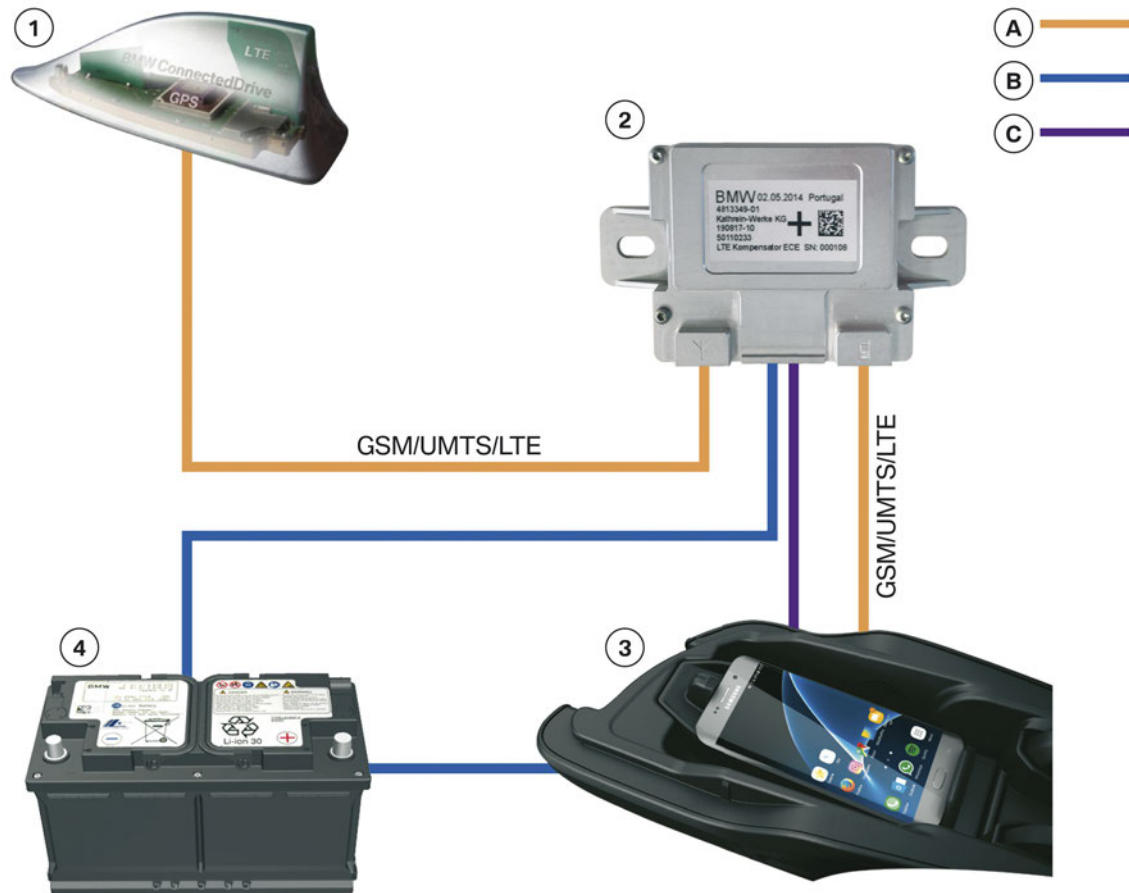
Index	Explanation
1	Mobile phone with coupling antenna for mobile reception
2	Induction coil of mobile phone
3	Outer cover of the mobile phone with receiver coil
4	Charging/transmitting coil in the wireless charging tray
5	Telephone antenna in the wireless charging station (inductive)
6	Ferrite in the wireless charging tray
7	Electronics with connections for the vehicle electrical system of the wireless charging tray
8	Outer cover of the wireless charging tray

The wireless charging function uses induction. A transmitter coil and a receiver coil contactlessly transmit the charging voltage for the mobile phone. At the same time, the smartphone is paired with the roof-mounted antenna.

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Connection to the vehicle electrical system



TE17-1103

F39 Connection to the vehicle electrical system

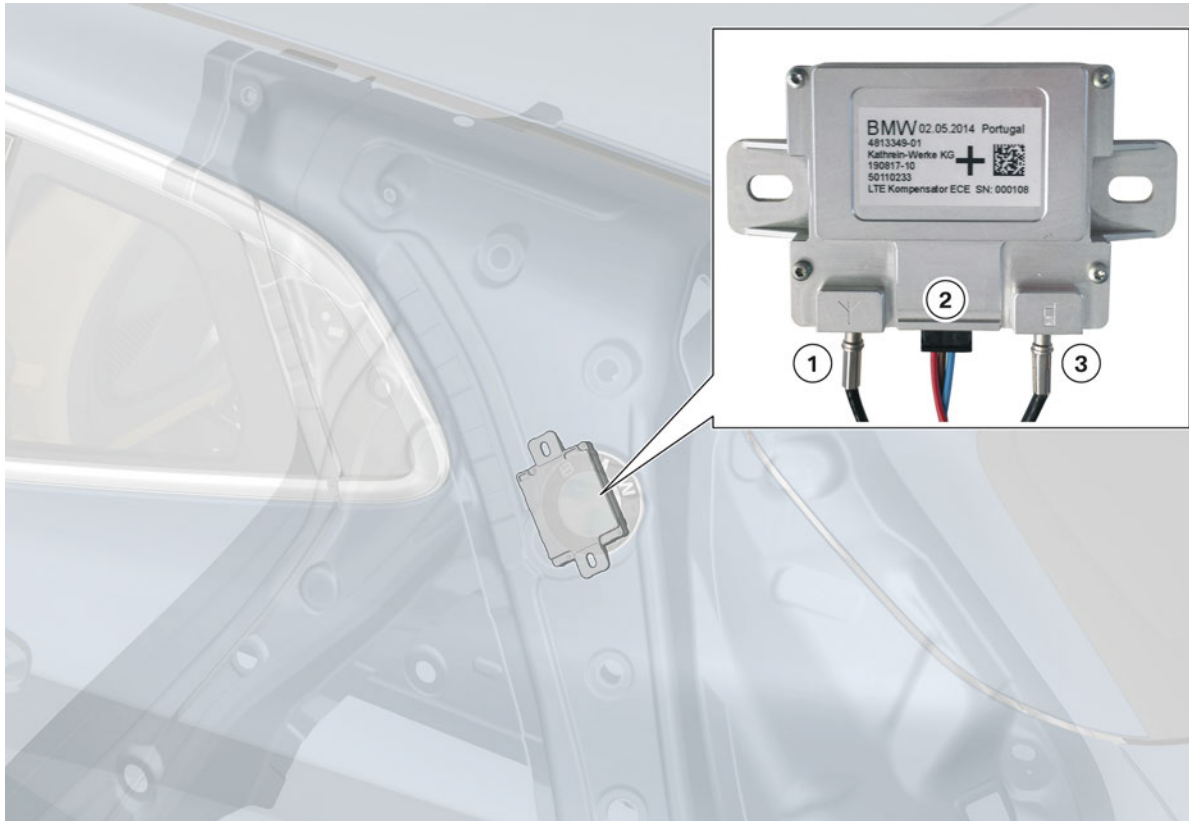
Index	Explanation
A	Antenna lines
B	Voltage supply
C	Activation line
1	TCB2/fin with telephone antenna TEL1
2	Power compensator
3	Wireless charging station
4	Vehicle electrical system voltage supply

The graphic shows a simplified version of how wireless charging is integrated in the vehicle electrical system.

A LTE-capable power compensator compensates for the losses in wireless transmission. With this power compensator, the reception quality of the wireless charging transmit/receive system again reaches the level of a conventional phone with a snap-in adapter.

F39 Complete Vehicle

10. Infotainment



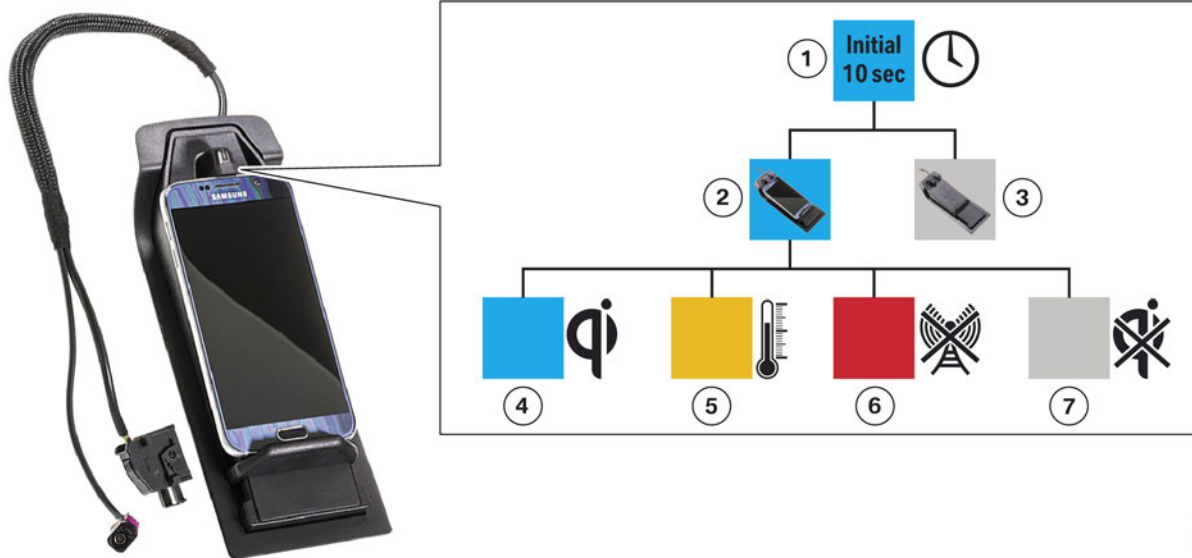
F39 Power compensator

Index	Explanation
1	Connection to the telephone antenna TEL1 in the TCB2 or roof fin antenna
2	Voltage supply, activation line of power compensator
3	Antenna connection of the wireless charging station

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Status LED



TE16-1229

Wireless charging status LED

Index	Explanation
1	Self-diagnosis
2	Check for mobile phone compatibility
3	Not possible to charge (e.g. mobile phone not compatible)
4	Mobile phone is charging
5	Charging not possible (e.g. temperature of the charging electronics in the mobile phone is too high or foreign bodies detected)
6	Charging not possible (fault with the antenna system)
7	Not possible to charge (e.g. mobile phone fully charged)

The status of the wireless charging tray is indicated by an LED.

The status LED turns blue as soon as the wireless charging station is activated by pulling the slide. During this time the electronics check whether the inserted object complies with the Qi® standard. If so, the status LED continues to be blue. If not, the status LED goes out.

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10. Infotainment

Because of self-diagnosis during activation of the wireless charging station (actuate slide and hold) even when no object has been inserted, the status LED shows blue. In this case, the status LED goes out after 10 s. The initial illumination of the LED may lead to customer misunderstandings.



The charging electronics heats up if there is a Qi[®]-compatible device and another metallic object (e.g. a coin) in the wireless charging station. The charging procedure is usually cancelled after a short time.



The Qi[®] standard is not identical in all devices. In certain mobile phones problems during the charging procedure can therefore occur.



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